Ex-14
2/10/2026

PACKET SNIFFING USING RAW SOCKETS

AIM:-

To implement packet sniffing using sockets to capture and identify networks packets and their source and destination IP addresses.

CODE:-

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
  if IP in packet:
    ip_layer = packet[IP]
    protocol = ip_layer.proto
    src_ip = ip_layer.src
    dst_ip = ip_layer.dst

    protocol_name = ""
    if protocol == 1:
      protocol_name = "ICMP"
    elif protocol == 6:
      protocol_name = "TCP"
    elif protocol == 17:
      protocol_name = "UDP"
    else:
      protocol_name = "Unknown Protocol"
```

```
    print(f"Protocol: {protocol_name}")
    print(f"Source IP: {src_ip}")
    print(f"Destination IP: {dst_ip}")
    print("-"*50)
sniff(iface='Wi-Fi', prn=packet_callback, filter="ip", store=0)
```

OUTPUT:-

```
C:\Users\tkala\Downloads> python packet.py
Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6
--------------------------------------------------
Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117
--------------------------------------------------
Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117
--------------------------------------------------
Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117
--------------------------------------------------
```

Protocol : TCP

Source IP : 57.155.141.117

Destination IP : 192.168.1.6

Protocol : TCP

Source IP : 57.155.141.117

Destination IP : 192.168.1.6

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

Protocol : UDP

Source IP : 192.168.1.1

Destination IP : 192.168.1.6

RESULT:-

Thus network packets were captured and their protocols and source and destination IP addresses were displayed successfully.